

Ujjwal Thapliyal

Data Scientist

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PROFILE

Results-driven Data Scientist with hands-on experience collecting, preprocessing, and analyzing large, complex datasets to generate actionable insights that drive business growth. Proficient in Python, SQL, and statistical modeling with a strong foundation in mathematics, machine learning, NLP, and end-to-end data projects — from ideation to production deployment. Skilled at presenting data-driven recommendations through compelling data visualizations and reports tailored for both technical and non-technical stakeholders. Experienced collaborating in remote, cross-functional team environments with a meticulous commitment to data quality and data integrity at every stage of the analytical workflow.

SKILLS

Programming Languages: Python, SQL, SciPy, Pandas, NumPy

Statistics & Mathematics: Probability, Inferential Statistics, Hypothesis Testing, Regression Analysis, Optimization, RMSE, MAE, Precision, Recall, F1-score

Machine Learning & Deep Learning: Classification, Regression, Clustering, NLP, CNN, RNN, Hyperparameter Optimization, Predictive Modeling, Model Validation

Data Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn, Analytical Dashboards (technical & non-technical audiences)

Data Handling & EDA: Data Collection, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Data Integrity, Data Quality, Insight Generation

Tools & Platforms: Scikit-learn, XGBoost, MLflow, FastAPI, Docker, Streamlit, Hugging Face

Soft Skills: Written and verbal communication, data-driven recommendations, cross-functional collaboration, remote team collaboration, attention to detail

PROFESSIONAL EXPERIENCE

AI ML Intern | TechnicalOne (Remote)

Jan 2026 – Feb 2026

- **Engineered an end-to-end NLP pipeline** for Fake News Detection using DistilBERT, ensuring data integrity and preprocessing quality across large text datasets — optimized for high classification accuracy and low inference latency.
- Collected, preprocessed, and tokenized large, complex text datasets to fine-tune a pre-trained transformer model, maintaining **data quality** at every stage to enable real-time misinformation detection.
- Deployed the fine-tuned model to production as an interactive web application on Hugging Face Spaces, communicating findings through a scalable interface accessible to end-users — demonstrating ability to deliver **end-to-end data projects from ideation to delivery**.

PROJECTS

Book Recommendation System | TF-IDF | K-Means | Cosine Similarity

Jan 2025 – Jan 2025

- Built a content-based recommendation system on **5,000+ items** using TF-IDF and cosine similarity; applied clustering techniques to improve recommendation diversity, relevance, and business value.
- Automated the ranking and retrieval pipeline, delivering personalized results with **sub-second response times** and reducing user search effort by ~40% — demonstrating focus on analytical methodology improvement and automation of data workflows.

Flight Price & Customer Satisfaction Prediction System , Machine Learning | Python | Scikit-learn | XGBoost | MLflow | Streamlit

Jun 2025 – Jul 2025

- Collected, cleaned, and preprocessed **110K+ flight and passenger records**, ensuring data integrity and readiness for analysis to support dynamic pricing strategy — identifying trends, patterns, and opportunities for business improvement.
- Conducted **exploratory data analysis (EDA)** to identify service quality drivers and pricing patterns; applied feature engineering to improve signal quality and model stability.
- Developed, validated, and implemented statistical models (Linear Regression, Random Forest, XGBoost) achieving **$R^2 = 0.85$** and reducing RMSE by **23% vs baseline**; rigorous evaluation via RMSE, MAE, precision, recall.
- Built classification models on 10K+ labeled samples achieving **80% accuracy and 0.80 F1-score**; delivered analytical dashboards with real-time performance monitoring for clear, impactful **data-driven recommendations** to both technical and non-technical stakeholders.
- Continuously improved analytical methodologies and automation processes throughout the project lifecycle, reducing manual analysis effort and enhancing data workflow efficiency.

FlipSatisfy – Customer Satisfaction Prediction | NLP | FastAPI | Docker

Jul 2025 – Aug 2025

- Designed a machine learning pipeline on **85,907 customer support records**, collecting and preprocessing diverse, unstructured data and categorical features into analysis-ready inputs — with strict attention to data quality and data integrity.
- Engineered NLP preprocessing workflows (TF-IDF vectorization, encoding) to extract actionable insights from complex text data; trained and benchmarked multiple algorithms improving test accuracy from **95.6% to 96.9%** through hyperparameter optimization.
- Deployed the optimized predictive model as a containerized **FastAPI production service** enabling real-time inference; integrated a visual reporting layer to present data-driven recommendations to operational stakeholders.

EDUCATION

Master of Computer Applications (MCA), SGRR University Dehradun

Jul 2024 – Present

Master Data Science Program, IIT Madras Guvi HCL

Nov 2025 – Jun 2025

Bachelor's Degree 2021–2024, SGRR University Dehradun

Jul 2021 – Jul 2024

CERTIFICATES

- Data Science Certification
Guvi HCL : IIT Madras 
- Power Bi Certificate -Guvi HCL